

Prioritized Planning for Four Franka Arms

Implementation and validation summary for the shared-table environment. Planning and collision decisions use cuRobo and GPU sphere checks; PyBullet is restricted to final mesh visualization.

FLOWEBRRT	VANILLA RRT	COLLISION MODEL
3/3 solved mean successful time 53.10 s median 44.86 s	1/3 solved successful time 46.25 s	295 spheres/arm 1,180 total spheres all saved paths audited

Benchmark results

Problem	FlowEBRRT	VanillaRRT	GPU replay audit
0: dense_opposing_weave	solved - 42.96 s	failed - 120.02 s	pass
1: center_lane_exchange	solved - 71.47 s	solved - 46.25 s	pass
2: diagonal_crossing	solved - 44.86 s	failed - 120.03 s	pass

The 120-second budget is shared across priority generations. Failed Vanilla runs retain their best valid partial paths. All six saved path sets passed the final cuRobo replay audit: **yes**.

What was added

- Geometry binding.** Every robot uses the same Panda URDF mesh and the same 295-sphere MorphIt/cuRobo model. The scene manifest verifies the URDF match and records 1,180 planning spheres.
- World/base transforms.** cuRobo produces spheres in each panda_link0 frame. The adapter applies $p_{world} = R_{world_base} * p_{base} + t_{world_base}$. Table and fixture poses use the inverse transform before entering each arm's local cuRobo scene.
- Articulated dynamic obstacles.** A completed higher-priority path is cached on the GPU as [time, 295, x-y-z-radius]. Lower-priority rollout waypoints are aligned by time and checked with batched sphere-pair tests. A finished path holds its final configuration indefinitely.
- Generic orchestration hook.** prioritized_planning.py now accepts an optional dynamic-obstacle adapter. Its original 2D/3D radius behavior is unchanged when the hook is absent.

Planning sequence

Generation	Robot	Collision constraints
0	NW	self + table/fixtures
1	NE	self/static + NW(t)
2	SW	self/static + NW(t) + NE(t)
3	SE	self/static + NW(t) + NE(t) + SW(t)

Correctness checks and discovered fix

The first post-run GPU replay exposed a root-goal edge case in scenario 2: SW started inside its goal, so inherited RRT accepted it immediately without checking future parking. The root acceptance rule now calls the same future-collision test as every other goal. Scenario 2 FlowEBRRT was rerun after this fix and passed the replay audit.

Validation completed with 12 focused tests, CUDA endpoint checks for all three scenarios, real dynamic-sphere conflict detection, byte-compilation, and a second synchronized cuRobo replay of every saved path. PyBullet was not imported by the planning or audit entry points.

Files and outputs

File	Purpose
franka_prioritized_adapter.py	Base transforms, cuRobo scenes, GPU sphere caches, dynamic and parking checks
prioritized_four_franka.py	Vanilla/Flow runner, path persistence, per-generation statistics
audit_four_franka_prioritized.py	Independent cuRobo-only synchronous path replay
visualize_four_franka_prioritized.py	PyBullet mesh playback and MP4 generation only
prioritized_planning.py	Backward-compatible optional articulated-obstacle hook
run_20260807_final	Six result folders, JSON/CSV statistics, audits, paths, and six videos

Complete per-generation data is stored in combined_generations.csv and in each result folder's generations.csv/summary.json. Videos are 640x480, 24 FPS, and contain 96 frames each.